AI Assisted Coding Assignment- 13

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Lab 13: Code Refactoring – Improving Legacy Code with AI Suggestions

Lab Objectives:

- Identify code smells and inefficiencies in legacy Python scripts.
- Use AI-assisted coding tools to **refactor** for readability, maintainability, and performance.
- Apply **modern Python best practices** while ensuring output correctness.

Task Description #1 (Refactoring – Removing Code Duplication)

- Task: Use AI to refactor a given Python script that contains multiple repeated code blocks.
- Instructions:
    - Prompt AI to identify duplicate logic and replace it with functions or classes.
    - Ensure the refactored code maintains the same output.
    - Add docstrings to all functions.
- Sample Legacy Code:

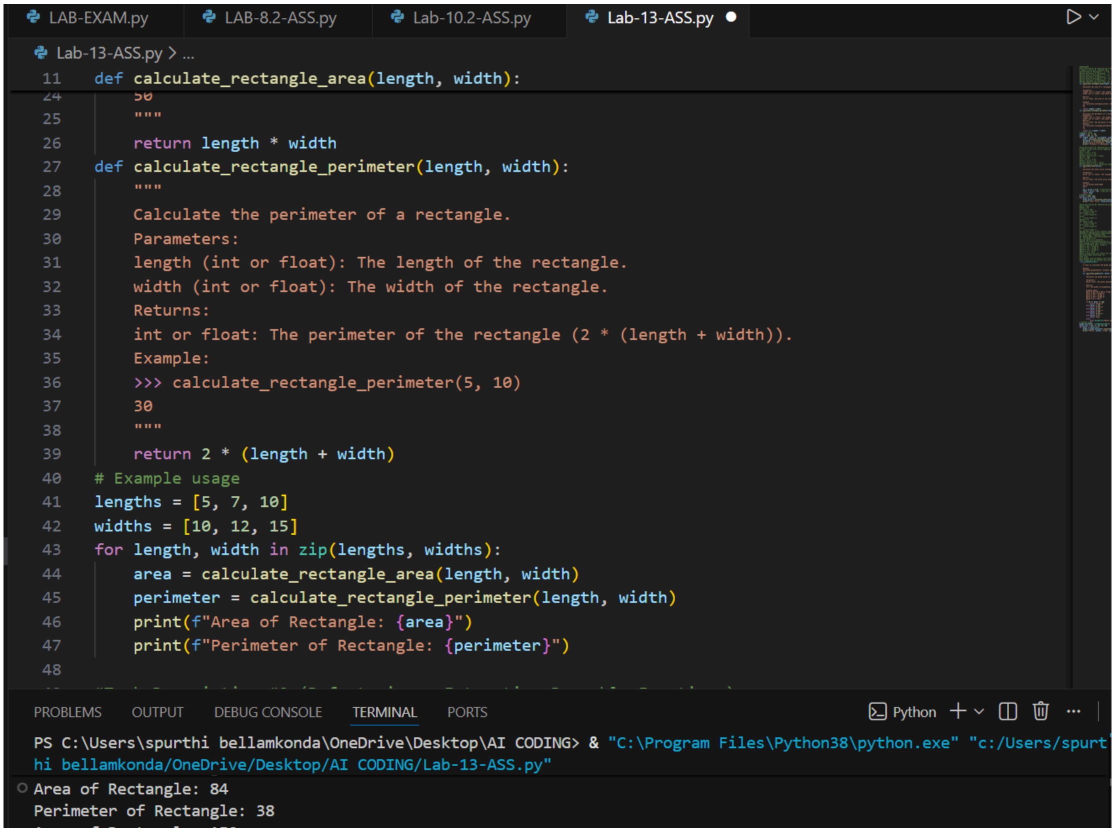
```
# Legacy script with repeated logic
print("Area of Rectangle:", 5 * 10)
print("Perimeter of Rectangle:", 2 * (5 + 10))
print("Area of Rectangle:", 7 * 12)
print("Perimeter of Rectangle:", 2 * (7 + 12))
print("Area of Rectangle:", 10 * 15)
print("Perimeter of Rectangle:", 2 * (10 + 15))
```

- Expected Output:
    - Refactored code with a reusable function and no duplication.
    - Well documented code

Lab-13-ASS.py > ...

```python
#Task Description #1 (Refactoring – Removing Code Duplication)
# Legacy script with repeated logic
#print("Area of Rectangle:", 5 * 10)
#print("Perimeter of Rectangle:", 2 * (5 + 10))
#print("Area of Rectangle:", 7 * 12)
#print("Perimeter of Rectangle:", 2 * (7 + 12))
#print("Area of Rectangle:", 10 * 15)
#print("Perimeter of Rectangle:", 2 * (10 + 15))
#Prompt:Identify duplicate logic and replace it with functions or classes.o Add docstrings to all fur
def calculate_rectangle_area(length, width):
    """
    Calculate the area of a rectangle.

    Parameters:
    length (int or float): The length of the rectangle.
    width (int or float): The width of the rectangle.

    Returns:
    int or float: The area of the rectangle (length * width).

    Example:
    >>> calculate_rectangle_area(5, 10)
    50
    """
    return length * width
def calculate_rectangle_perimeter(length, width):
    """
    Calculate the perimeter of a rectangle.
    Parameters:
    length (int or float): The length of the rectangle.
    width (int or float): The width of the rectangle.
    Returns:
    int or float: The perimeter of the rectangle (2 * (length + width)).
    Example:
    >>> calculate_rectangle_perimeter(5, 10)
    30
```

Ln 6, Col 37 Spaces: 4 UTF-8 CRLF { } Python Python 3.8 (64-bit) Go Live



```
LAB-EXAM.py    LAB-8.2-ASS.py    Lab-10.2-ASS.py    Lab-13-ASS.py
Lab-13-ASS.py > ...
def calculate_rectangle_area(length, width):
    50
    """
    return length * width
def calculate_rectangle_perimeter(length, width):
    """
    Calculate the perimeter of a rectangle.
    Parameters:
    length (int or float): The length of the rectangle.
    width (int or float): The width of the rectangle.
    Returns:
    int or float: The perimeter of the rectangle (2 * (length + width)).
    Example:
    >>> calculate_rectangle_perimeter(5, 10)
    30
    """
    return 2 * (length + width)
# Example usage
lengths = [5, 7, 10]
widths = [10, 12, 15]
for length, width in zip(lengths, widths):
    area = calculate_rectangle_area(length, width)
    perimeter = calculate_rectangle_perimeter(length, width)
    print(f"Area of Rectangle: {area}")
    print(f"Perimeter of Rectangle: {perimeter}")
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> & "C:\Program Files\Python38\python.exe" "c:/Users/spurthi bellamkonda/OneDrive/Desktop/AI CODING/Lab-13-ASS.py"
Area of Rectangle: 84
Perimeter of Rectangle: 38
```

Task Description #2 (Refactoring – Extracting Reusable Functions)

- Task: Use AI to refactor a legacy script where multiple calculations are embedded directly inside the main code block.
- Instructions:
  - Identify repeated or related logic and extract it into reusable functions.
  - Ensure the refactored code is modular, easy to read, and documented with docstrings.
- Sample Legacy Code:

Legacy script with inline repeated logic

price = 250

tax = price * 0.18

total = price + tax

print("Total Price:", total)

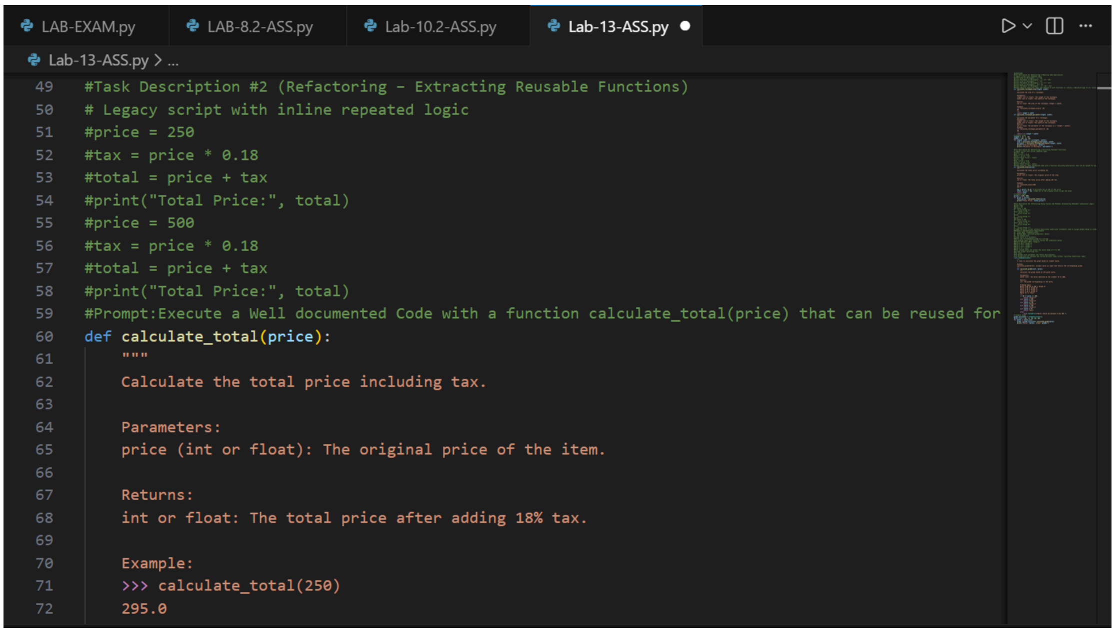
price = 500

tax = price * 0.18

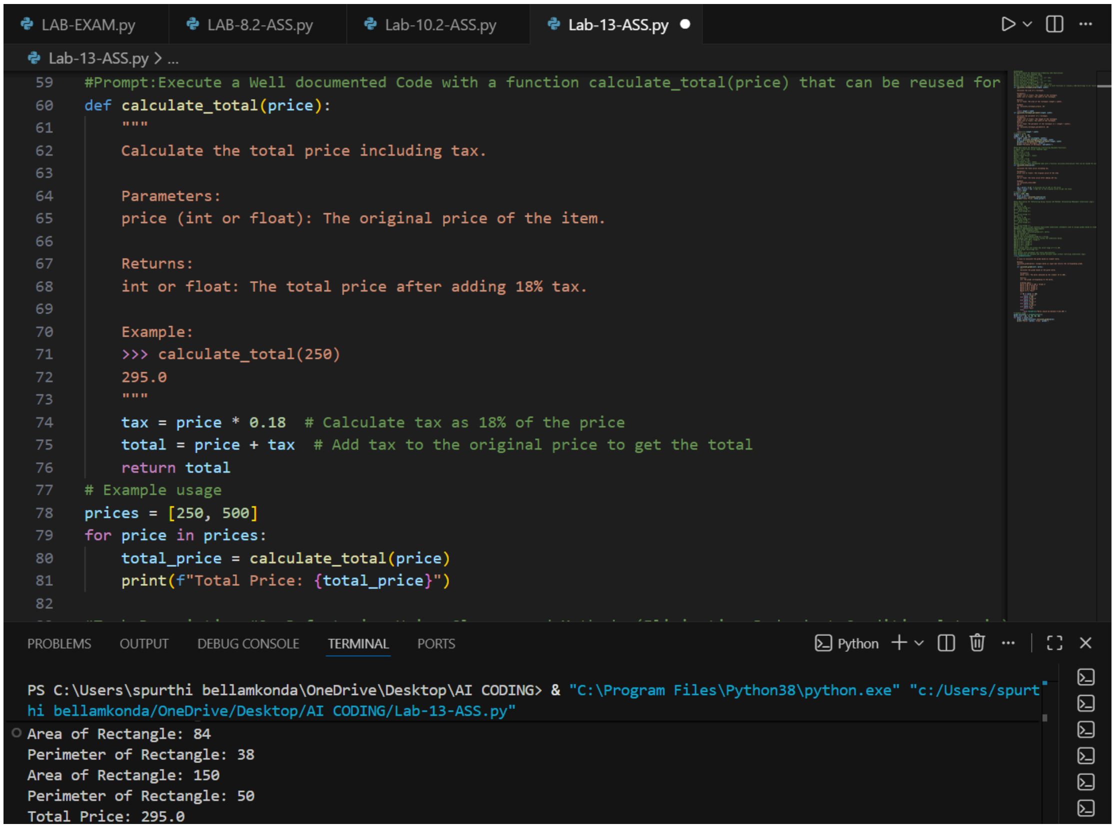
total = price + tax

print("Total Price:", total)

- Expected Output:
    - Code with a function calculate_total(price) that can be reused for multiple price inputs.
    - Well documented code




```python
#Task Description #2 (Refactoring – Extracting Reusable Functions)
# Legacy script with inline repeated logic
#price = 250
#tax = price * 0.18
#total = price + tax
#print("Total Price:", total)
#price = 500
#tax = price * 0.18
#total = price + tax
#print("Total Price:", total)
#Prompt:Execute a Well documented Code with a function calculate_total(price) that can be reused for
def calculate_total(price):
    """
    Calculate the total price including tax.

    Parameters:
    price (int or float): The original price of the item.

    Returns:
    int or float: The total price after adding 18% tax.

    Example:
    >>> calculate_total(250)
    295.0
```



<u>Task Description #3:</u> Refactoring Using Classes and Methods (Eliminating Redundant Conditional Logic)

Refactor a Python script that contains repeated if–elif–else grading logic by implementing a structured, object-oriented solution using a class and a method.

Problem Statement

The given script contains duplicated conditional statements used to assign grades based on student marks. This redundancy violates clean code principles and reduces maintainability.

You are required to refactor the script using a class-based design to improve modularity, reusability, and readability while preserving the original grading logic.

Mandatory Implementation Requirements

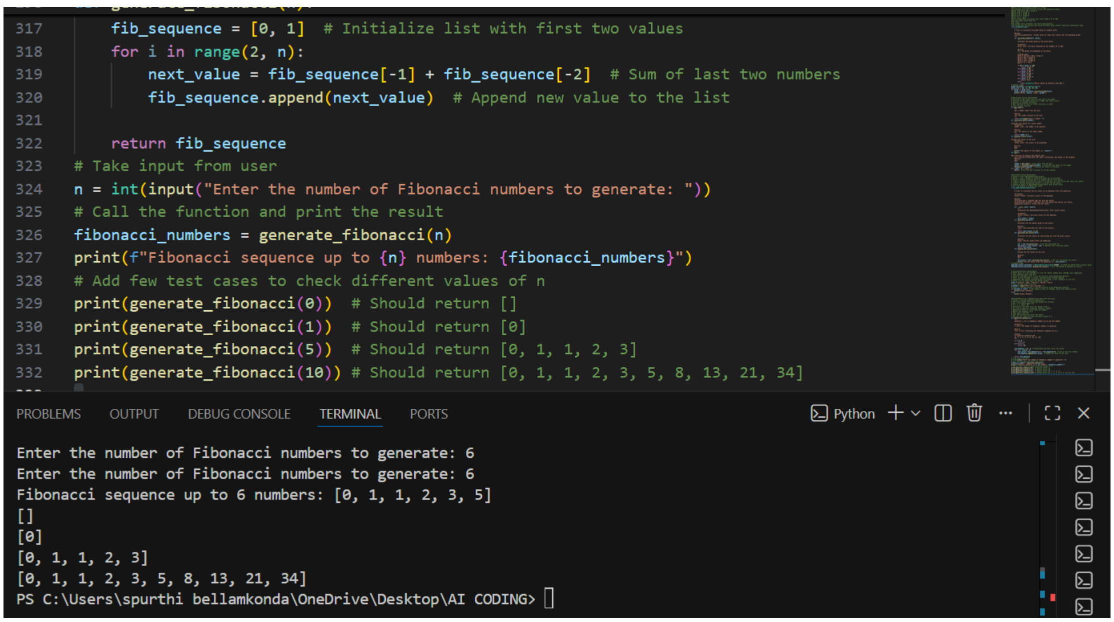
1. Class Name: GradeCalculator
2. Method Name: calculate_grade(self, marks)
3. The method must:
    - Accept marks as a parameter.
    - Return the corresponding grade as a string.

- The grading logic must strictly follow the conditions below:

- Marks ≥ 90 and ≤ 100 → "Grade A"
- Marks ≥ 80 → "Grade B"
- Marks ≥ 70 → "Grade C"
- Marks ≥ 40 → "Grade D"
- Marks ≥ 0 → "Fail"

Note: Assume marks are within the valid range of 0 to 100.

1. Include proper docstrings for:
    - The class
    - The method (with parameter and return descriptions)



2. The method must be reusable and called multiple times without rewriting conditional logic.

- Given code:

marks = 85

if marks >= 90:

print("Grade A")

elif marks >= 75:

print("Grade B")

else:

print("Grade C")

marks = 72

if marks >= 90:

print("Grade A")

elif marks >= 75:

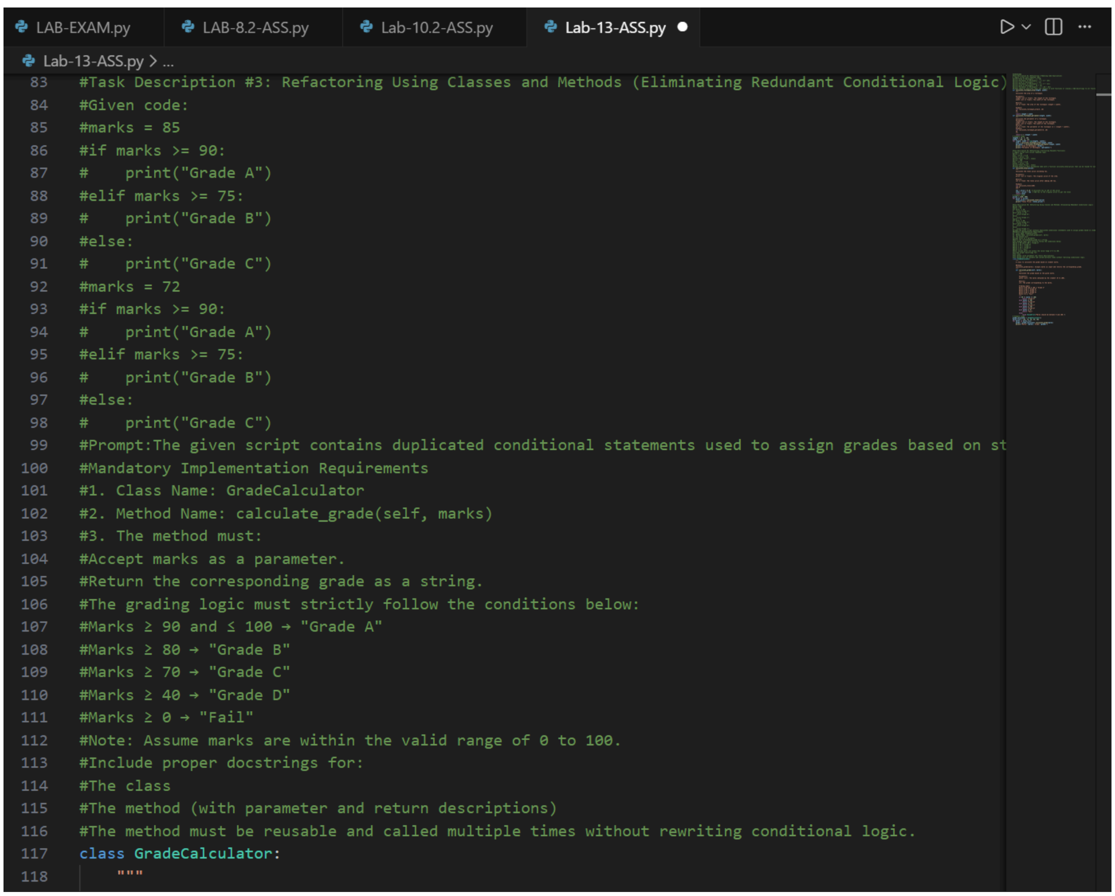
print("Grade B")

else:

print("Grade C")

Expected Output:

- Define a class named GradeCalculator.
- Implement a method calculate_grade(self, marks) inside the class.
- Create an object of the class.
- Call the method for different student marks.
- Print the returned grade values.

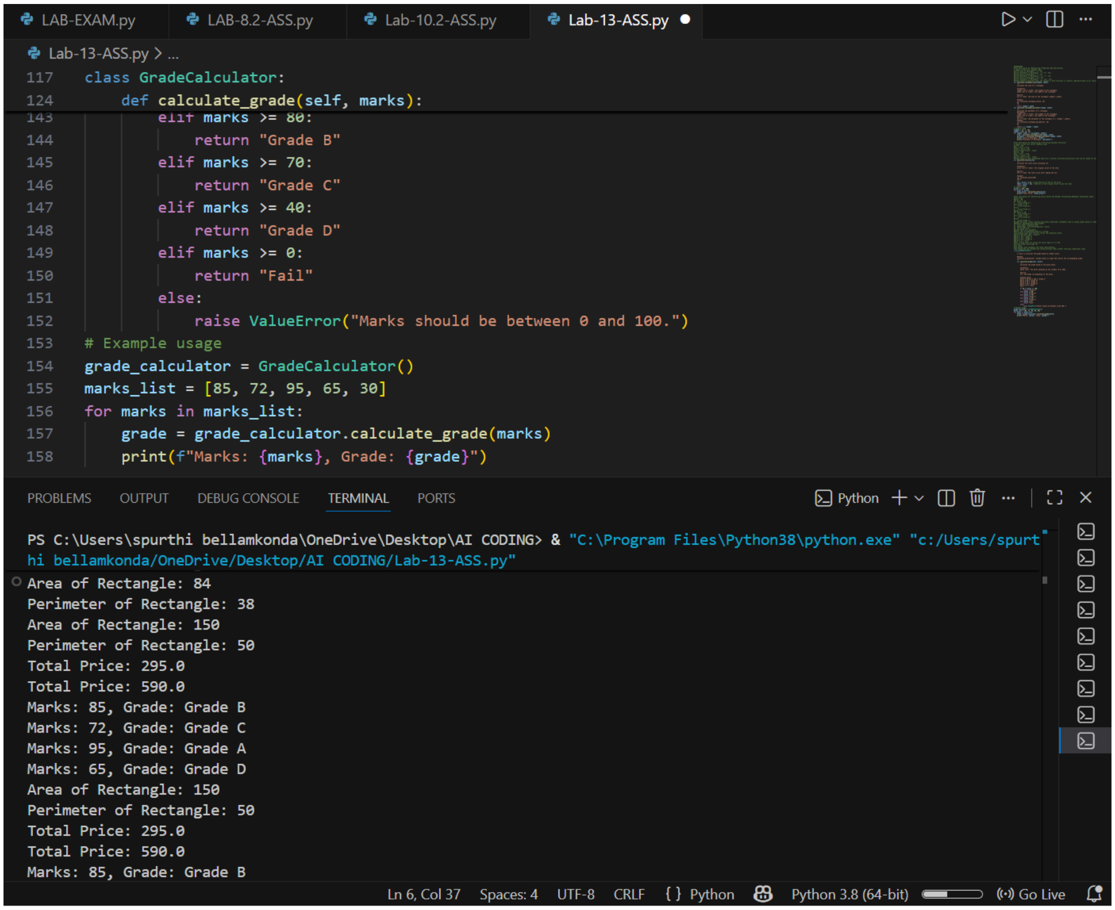


```
LAB-EXAM.py   LAB-8.2-ASS.py   Lab-10.2-ASS.py   Lab-13-ASS.py
Lab-13-ASS.py > ...
#Task Description #3: Refactoring Using Classes and Methods (Eliminating Redundant Conditional Logic)
#Given code:
#marks = 85
#if marks >= 90:
#    print("Grade A")
#elif marks >= 75:
#    print("Grade B")
#else:
#    print("Grade C")
#marks = 72
#if marks >= 90:
#    print("Grade A")
#elif marks >= 75:
#    print("Grade B")
#else:
#    print("Grade C")
#Prompt:The given script contains duplicated conditional statements used to assign grades based on st
#Mandatory Implementation Requirements
#1. Class Name: GradeCalculator
#2. Method Name: calculate_grade(self, marks)
#3. The method must:
#Accept marks as a parameter.
#Return the corresponding grade as a string.
#The grading logic must strictly follow the conditions below:
#Marks ≥ 90 and ≤ 100 → "Grade A"
#Marks ≥ 80 → "Grade B"
#Marks ≥ 70 → "Grade C"
#Marks ≥ 40 → "Grade D"
#Marks ≥ 0 → "Fail"
#Note: Assume marks are within the valid range of 0 to 100.
#Include proper docstrings for:
#The class
#The method (with parameter and return descriptions)
#The method must be reusable and called multiple times without rewriting conditional logic.
class GradeCalculator:
    """
```

LAB-EXAM.py | LAB-8.2-ASS.py | Lab-10.2-ASS.py | Lab-13-ASS.py

Lab-13-ASS.py > ...

```python
class GradeCalculator:
    Methods:
    calculate_grade(marks): Accepts marks as input and returns the corresponding grade.
    """
    def calculate_grade(self, marks):
        """
        Calculate the grade based on the given marks.

        Parameters:
        marks (int): The marks obtained by the student (0 to 100).

        Returns:
        str: The grade corresponding to the marks.

        Grading Logic:
        Marks ≥ 90 and ≤ 100 → "Grade A"
        Marks ≥ 80 → "Grade B"
        Marks ≥ 70 → "Grade C"
        Marks ≥ 40 → "Grade D"
        Marks ≥ 0 → "Fail"
        """
        if 90 <= marks <= 100:
            return "Grade A"
        elif marks >= 80:
            return "Grade B"
        elif marks >= 70:
            return "Grade C"
        elif marks >= 40:
            return "Grade D"
        elif marks >= 0:
            return "Fail"
        else:
            raise ValueError("Marks should be between 0 and 100.")
# Example usage
grade_calculator = GradeCalculator()
marks_list = [85, 72, 95, 65, 30]
```



Task Description #4 (Refactoring – Converting Procedural Code to Functions)

- **Task: Use AI to refactor procedural input–processing logic into functions.**

Instructions:

o Identify input, processing, and output sections.

o Convert each into a separate function.

o Improve code readability without changing behavior.

- Sample Legacy Code:

num = int(input("Enter number: "))

square = num * num

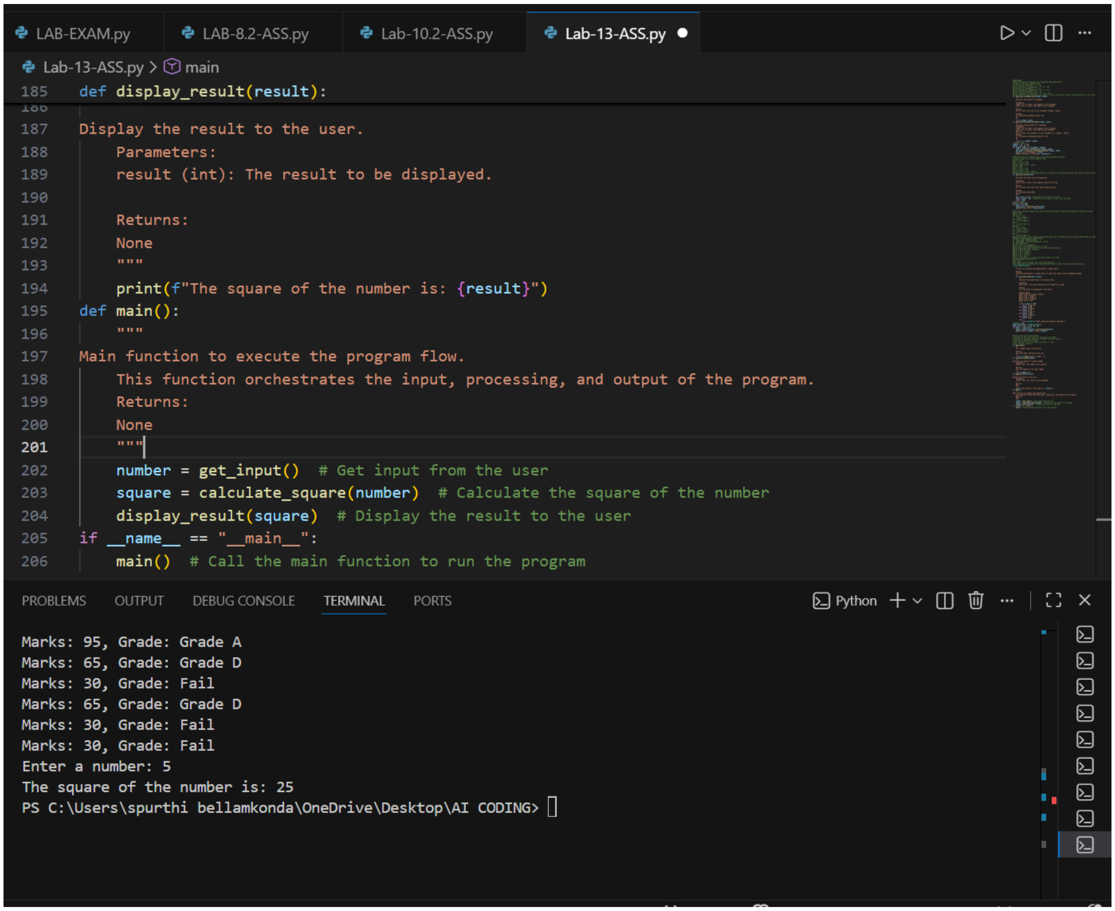
print("Square:", square)

- Expected Output:o Modular code using functions like get_input(), calculate_square(), and display_result().

LAB-EXAM.py LAB-8.2-ASS.py Lab-10.2-ASS.py Lab-13-ASS.py

Lab-13-ASS.py > ...

```python
#Task-4(Code Clarity Improvement)
# Function to get input from user and return the number
# Function to calculate square of a number and return result
# Function to display the result
# Main function to call all other functions in order
# Call the main function
def get_input():
    """
    Get a number input from the user.

    Returns:
    int: The number entered by the user.
    """
    return int(input("Enter a number: "))
def calculate_square(number):
    """
Calculate the square of a given number.
    Parameters:
    number (int): The number to be squared.

    Returns:
    int: The square of the input number.
    """
    return number ** 2
def display_result(result):
    """
Display the result to the user.
    Parameters:
    result (int): The result to be displayed.

    Returns:
    None
```



Task 5 (Refactoring Procedural Code into OOP Design)

- **Task: Use AI to refactor procedural code into a class-based design.Focus Areas:**o Object-Oriented principleso Encapsulation

Legacy Code:

salary = 50000

tax = salary * 0.2

net = salary - tax

print(net)

Expected Outcome:

o A class like EmployeeSalaryCalculator with methods and attributes.

```python
#Task-5(Efficiency Enhancement)
# Create a class EmployeeSalaryCalculator
# Add an __init__ method to initialize salary as an attribute
# Create a method calculate_tax() to compute 20% tax from salary
# Create a method calculate_net_salary() to subtract tax from salary and return net salary
# Create a method display_net_salary() to print the net salary
# Create an object of EmployeeSalaryCalculator with salary value
# Call the method to display net salary
class EmployeeSalaryCalculator:
    """
    A class to calculate the net salary of an employee after tax deduction.

    Attributes:
    salary (float): The gross salary of the employee.

    Methods:
    calculate_tax(): Computes 20% tax from the salary.
    calculate_net_salary(): Subtracts tax from salary and returns net salary.
    display_net_salary(): Prints the net salary.
    """
    def __init__(self, salary):
        """
        Initialize the EmployeeSalaryCalculator with a given salary.

        Parameters:
        salary (float): The gross salary of the employee.
        """
        self.salary = salary
    def calculate_tax(self):
        """
```

```python
class EmployeeSalaryCalculator:
        return self.salary * 0.20
    def calculate_net_salary(self):
        """
        Calculate the net salary by subtracting tax from the gross salary.

        Returns:
        float: The net salary after tax deduction.
        """
        tax = self.calculate_tax()  # Get the calculated tax
        net_salary = self.salary - tax  # Subtract tax from gross salary
        return net_salary
    def display_net_salary(self):
        """
        Display the net salary to the user.

        Returns:
        None
        """
        net_salary = self.calculate_net_salary()  # Get the calculated net salary
        print(f"The net salary after tax deduction is: {net_salary}")
# Example usage
employee_salary_calculator = EmployeeSalaryCalculator(50000)  # Create an object with a salary value
employee_salary_calculator.display_net_salary()  # Call the method to display net salary
```

```
Enter a number: 5
The square of the number is: 25
Enter a number: 5
The square of the number is: 25
The square of the number is: 25
The net salary after tax deduction is: 40000.0
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Task 6 (Optimizing Search Logic)

- **Task: Refactor inefficient linear searches using appropriate data structures.**Focus Areas:o Time complexityo Data structure choice

Legacy Code:

users = ["admin", "guest", "editor", "viewer"]

name = input("Enter username: ")

found = False

for u in users:

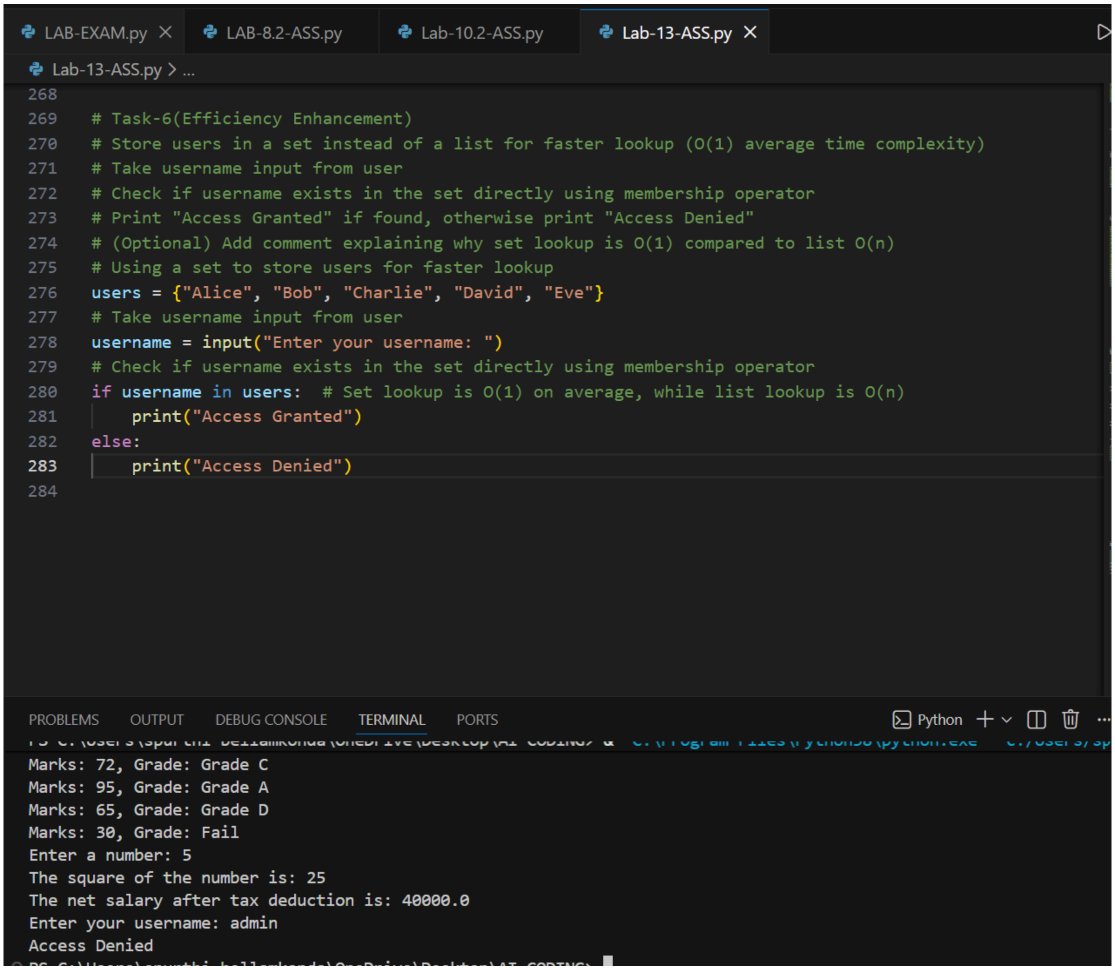
if u == name:

found = True

print("Access Granted" if found else "Access Denied")

Expected Outcome:

o Use of sets or dictionaries with complexity justification



<u>Task 7 – Refactoring the Library Management System</u>

Problem Statement

You are provided with a poorly structured Library Management script that:

- Contains repeated conditional logic
- Does not use reusable functions
- Lacks documentation
- Uses print-based procedural execution
- Does not follow modular programming principles

Your task is to refactor the code into a proper format

1. Create a module library.py with functions:

o add_book(title, author, isbn)

o remove_book(isbn)

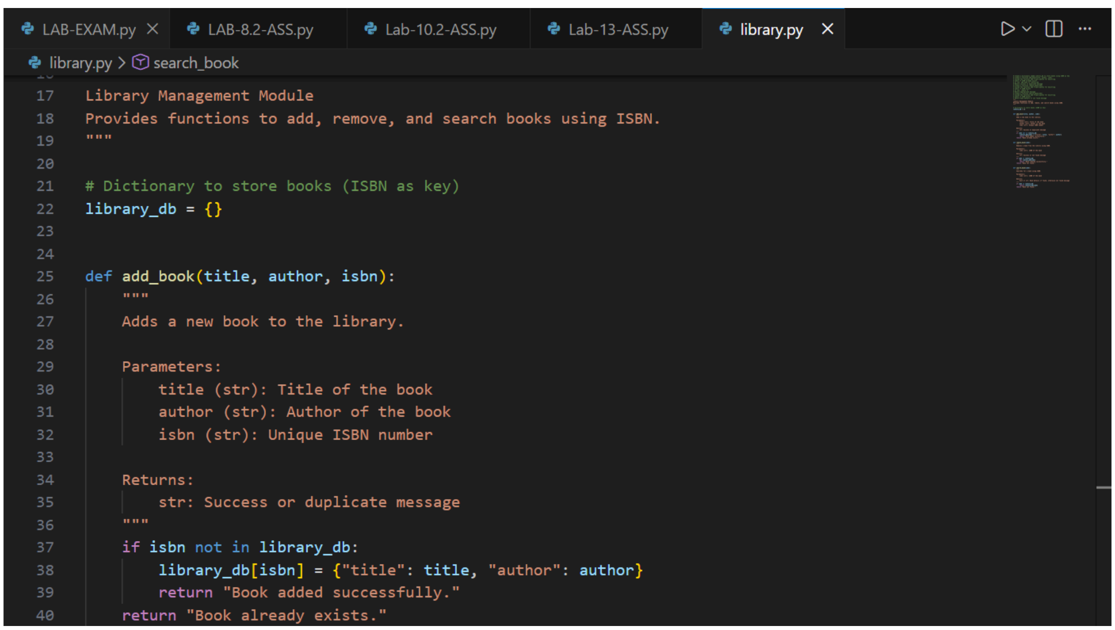
o search_book(isbn)

2. Insert triple quotes under each function and let Copilot complete the docstrings.
3. Generate documentation in the terminal.
4. Export the documentation in HTML format.
5. Open the file in a browser.

Given Code

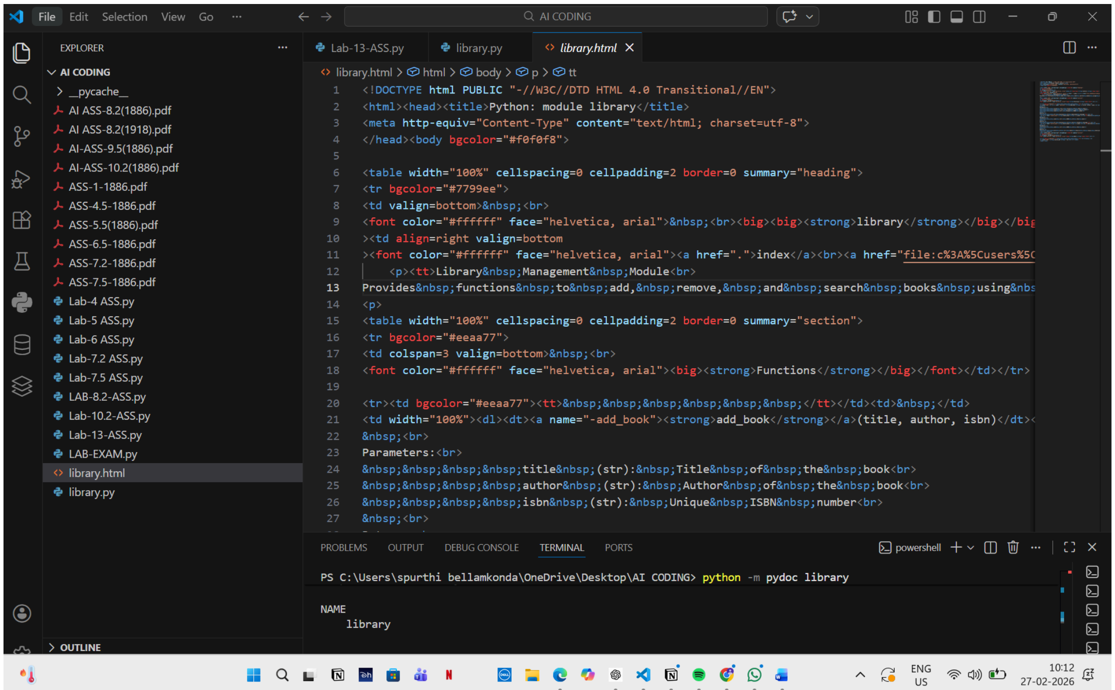
```
# Library Management System (Unstructured Version)
# This code needs refactoring into a proper module with documentation.
library_db = {}
# Adding first book
title = "Python Basics"
author = "John Doe"
isbn = "101"
if isbn not in library_db:
library_db[isbn] = {"title": title, "author": author}
print("Book added successfully.")
else:
print("Book already exists.")
# Adding second book (duplicate logic)
title = "AI Fundamentals"
author = "Jane Smith"
isbn = "102"
if isbn not in library_db:
library_db[isbn] = {"title": title, "author": author}
print("Book added successfully.")
else:
print("Book already exists.")
# Searching book (repeated logic structure)
isbn = "101"
if isbn in library_db:
print("Book Found:", library_db[isbn])
else:
print("Book not found.")
# Removing book (again repeated pattern)
isbn = "101"
```

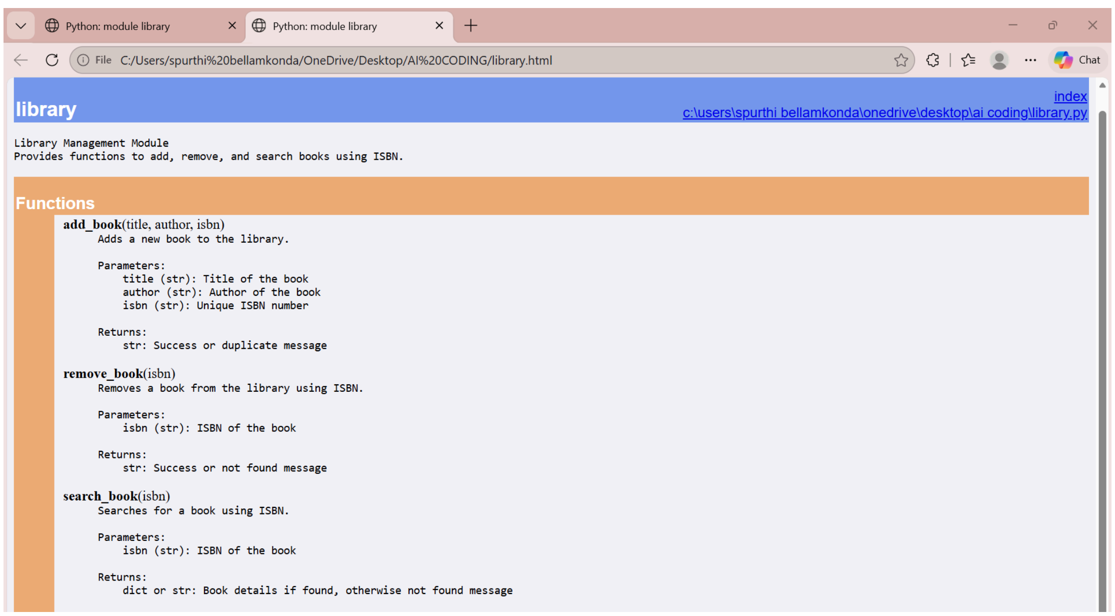
```
if isbn in library_db:
del library_db[isbn]
print("Book removed successfully.")
else:
print("Book not found.")
# Searching again
isbn = "101"
if isbn in library_db:
print("Book Found:", library_db[isbn])
else:
print("Book not found.")
```



```
LAB-EXAM.py   LAB-8.2-ASS.py   Lab-10.2-ASS.py   Lab-13-ASS.py   library.py
library.py > search_book
def remove_book(isbn):
        isbn (str): ISBN of the book

    Returns:
        str: Success or not found message
    """
    if isbn in library_db:
        del library_db[isbn]
        return "Book removed successfully."
    return "Book not found."


def search_book(isbn):
    """
    Searches for a book using ISBN.

    Parameters:
        isbn (str): ISBN of the book

    Returns:
        dict or str: Book details if found, otherwise not found message
    """
    if isbn in library_db:
        return library_db[isbn]
```

```
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS          powershell

PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> pydoc library

PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> python -m pydoc library
Help on module library:

NAME
    library


DESCRIPTION
    Library Management Module
    Provides functions to add, remove, and search books using ISBN.


PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> python -m pydoc -w library
wrote library.html
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING> start library.html
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```





Task 8– Fibonacci Generator

Write a program to generate Fibonacci series up to n.

The initial code has:

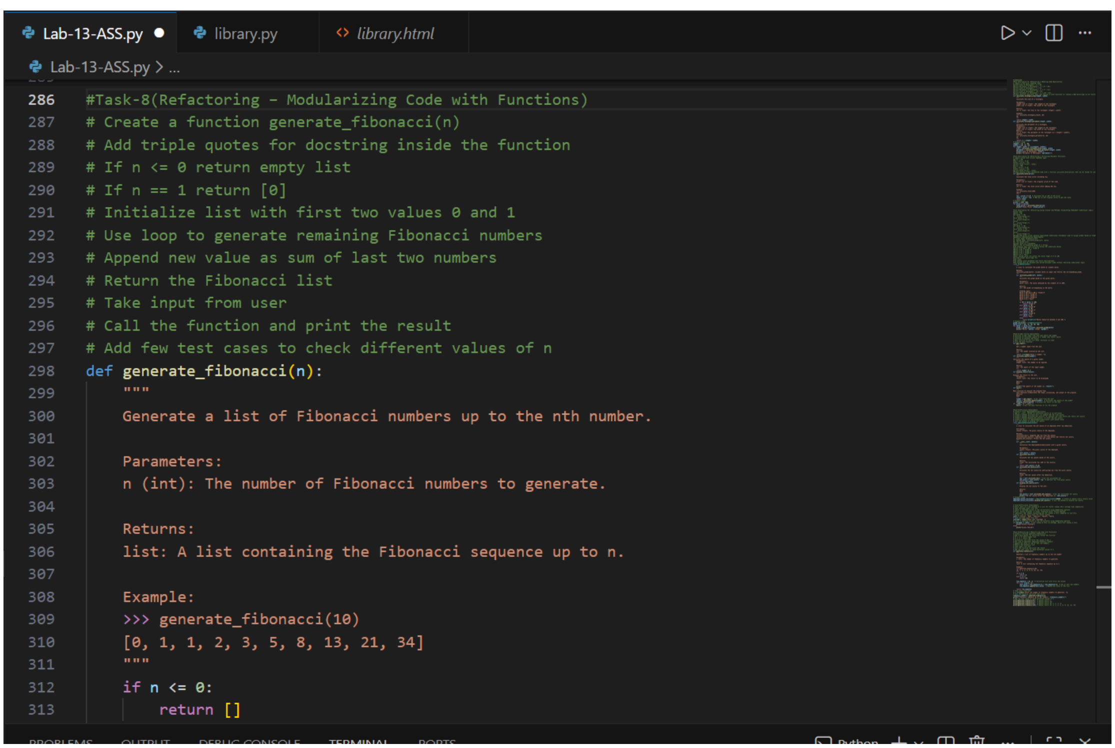
- Global variables.
- Inefficient loop.
- No functions or modularity.

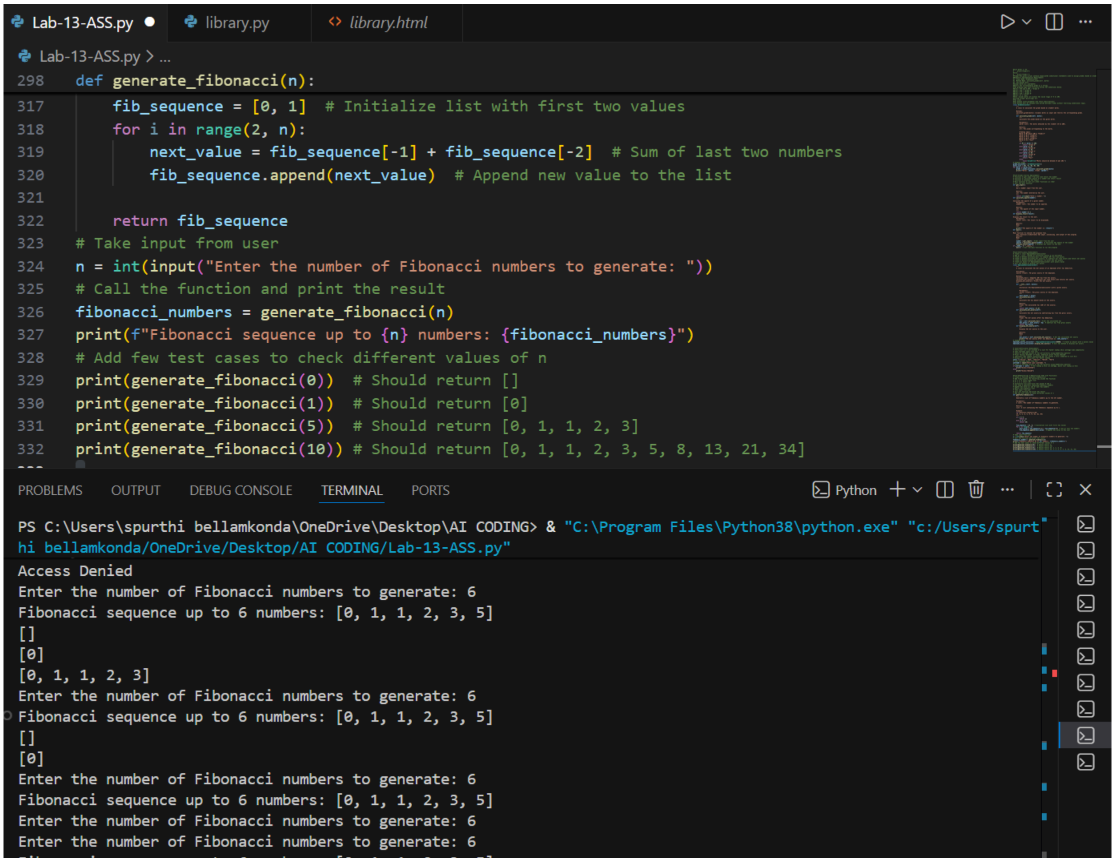
Task for Students:

- Refactor into a clean reusable function (generate_fibonacci).
- Add docstrings and test cases.
- Compare AI-refactored vs original.

Bad Code Version:

```
# fibonacci bad version
n=int(input("Enter limit: "))
a=0
b=1
print(a)
print(b)
for i in range(2,n):
c=a+b
print(c)
a=b
b=c
```





Task 9 – Twin Primes Checker

Twin primes are pairs of primes that differ by 2 (e.g., 11 and 13, 17 and 19).

The initial code has:

- Inefficient prime checking.
- No functions.
- Hardcoded inputs.

Task for Students:

- Refactor into is_prime(n) and is_twin_prime(p1, p2).
- Add docstrings and optimize.
- Generate a list of twin primes in a given range using AI.

Bad Code Version:

twin primes bad version

a=11

b=13

fa=0

for i in range(2,a):

if a%i==0:

fa=1

fb=0

for i in range(2,b):

if b%i==0:

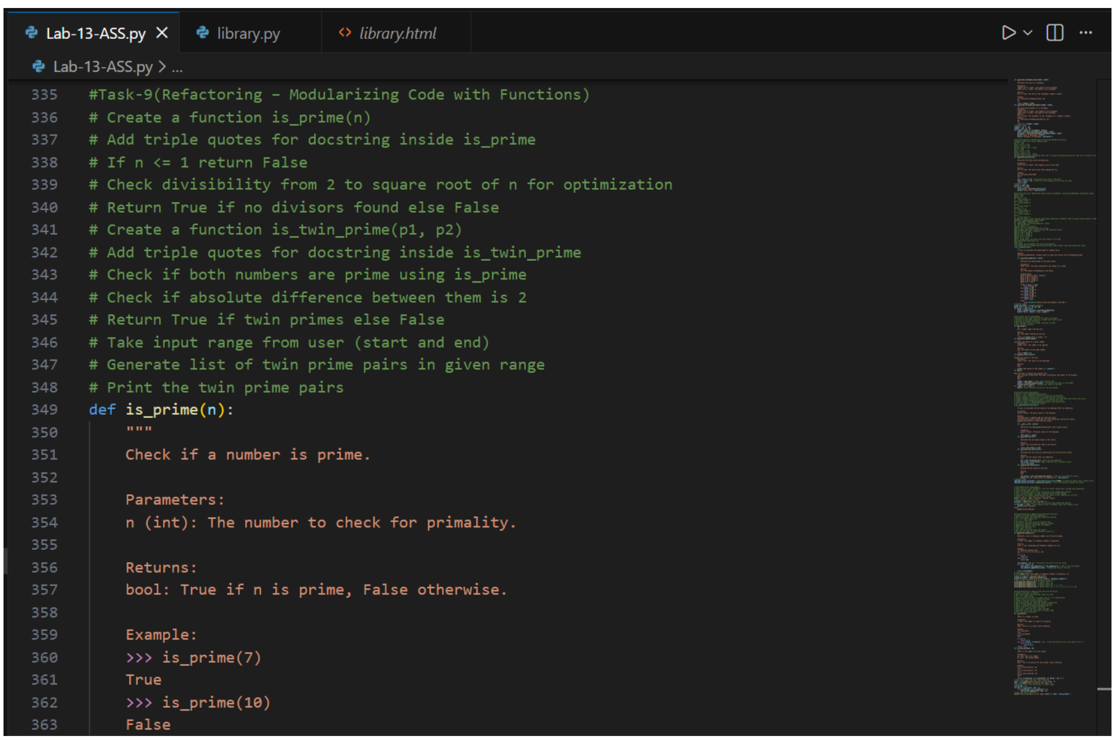
fb=1

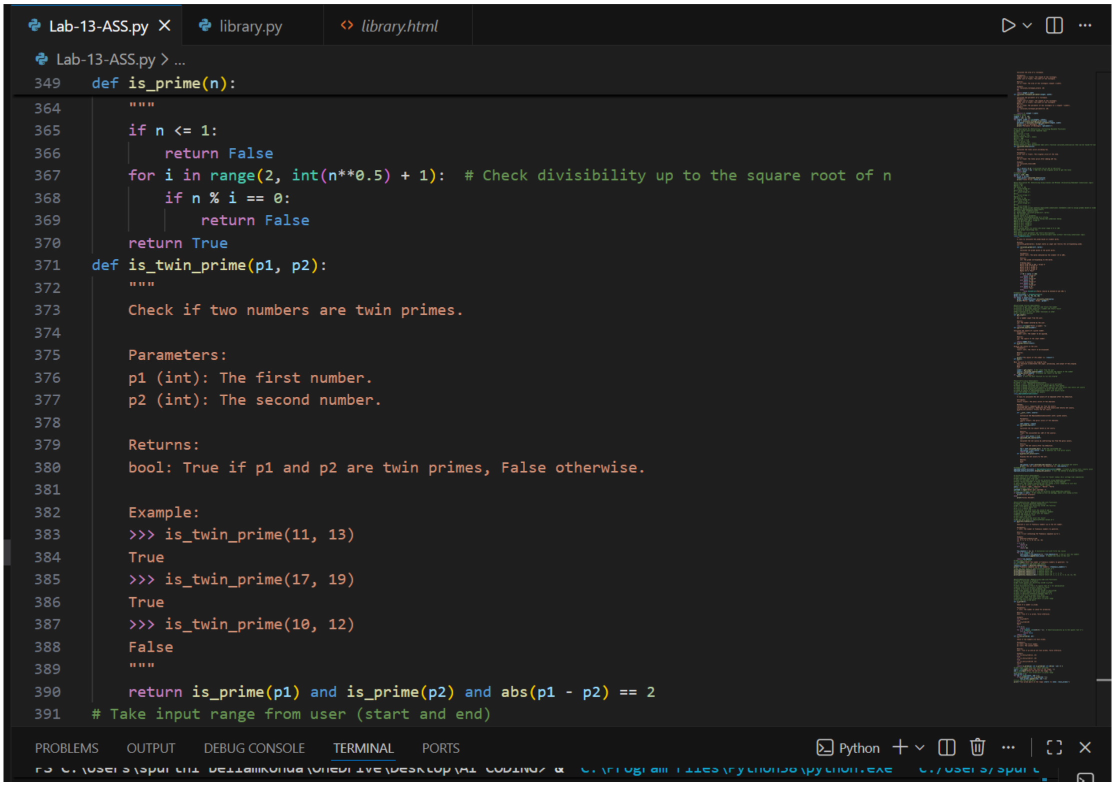
if fa==0 and fb==0 and abs(a-b)==2:

print("Twin Primes")

else:

print("Not Twin Primes")





```
Lab-13-ASS.py    library.py    library.html
Lab-13-ASS.py > ...
def is_prime(n):
    """
    if n <= 1:
        return False
    for i in range(2, int(n**0.5) + 1):  # Check divisibility up to the square root of n
        if n % i == 0:
            return False
    return True
def is_twin_prime(p1, p2):
    """
    Check if two numbers are twin primes.

    Parameters:
    p1 (int): The first number.
    p2 (int): The second number.

    Returns:
    bool: True if p1 and p2 are twin primes, False otherwise.

    Example:
    >>> is_twin_prime(11, 13)
    True
    >>> is_twin_prime(17, 19)
    True
    >>> is_twin_prime(10, 12)
    False
    """
    return is_prime(p1) and is_prime(p2) and abs(p1 - p2) == 2
# Take input range from user (start and end)
```

```
Lab-13-ASS.py    library.py    library.html
Lab-13-ASS.py > ...
def is_twin_prime(p1, p2):
    """
    return is_prime(p1) and is_prime(p2) and abs(p1 - p2) == 2
# Take input range from user (start and end)
start = int(input("Enter the start of the range: "))
end = int(input("Enter the end of the range: "))
# Generate list of twin prime pairs in given range
twin_primes = []
for num in range(start, end + 1):
    if is_prime(num) and is_prime(num + 2):
        twin_primes.append((num, num + 2))
# Print the twin prime pairs
print(f"Twin prime pairs in the range {start} to {end}: {twin_primes}")
```

```
[0]
[0, 1, 1, 2, 3]
[0, 1, 1, 2, 3, 5, 8, 13, 21, 34]
Enter the start of the range: 1
Enter the end of the range: 20
Twin prime pairs in the range 1 to 20: [(3, 5), (5, 7), (11, 13), (17, 19)]
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Task 10 – Refactoring the Chinese Zodiac Program

Objective

Refactor the given poorly structured Python script into a clean, modular, and reusable implementation.

The current program reads a year from the user and prints the corresponding Chinese Zodiac sign. However, the implementation contains repetitive conditional logic, lacks modular design, and does not follow clean coding principles.

Your task is to refactor the code to improve readability, maintainability, and structure.

Chinese Zodiac Cycle (Repeats Every 12 Years)

1. Rat
2. Ox
3. Tiger
4. Rabbit
5. Dragon
6. Snake
7. Horse
8. Goat (Sheep)
9. Monkey
10. Rooster
11. Dog
12. Pig

 # Chinese Zodiac Program (Unstructured Version)

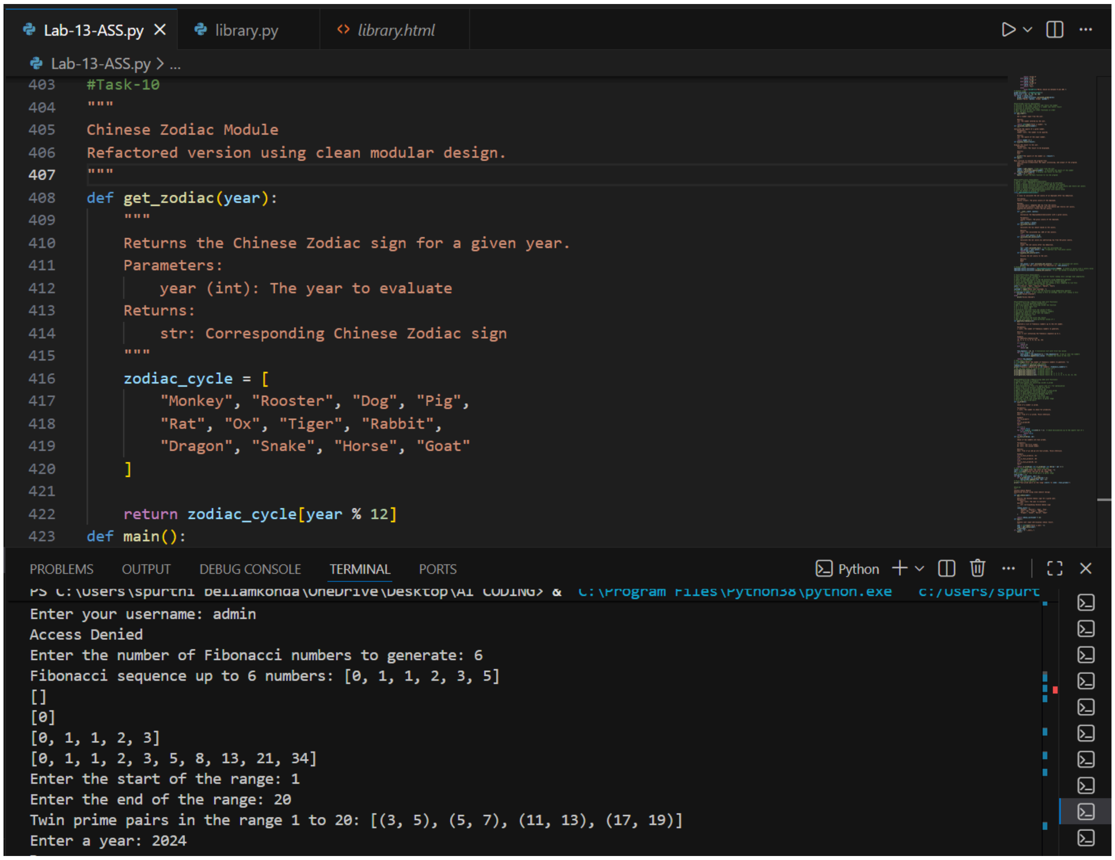
 # This code needs refactoring.

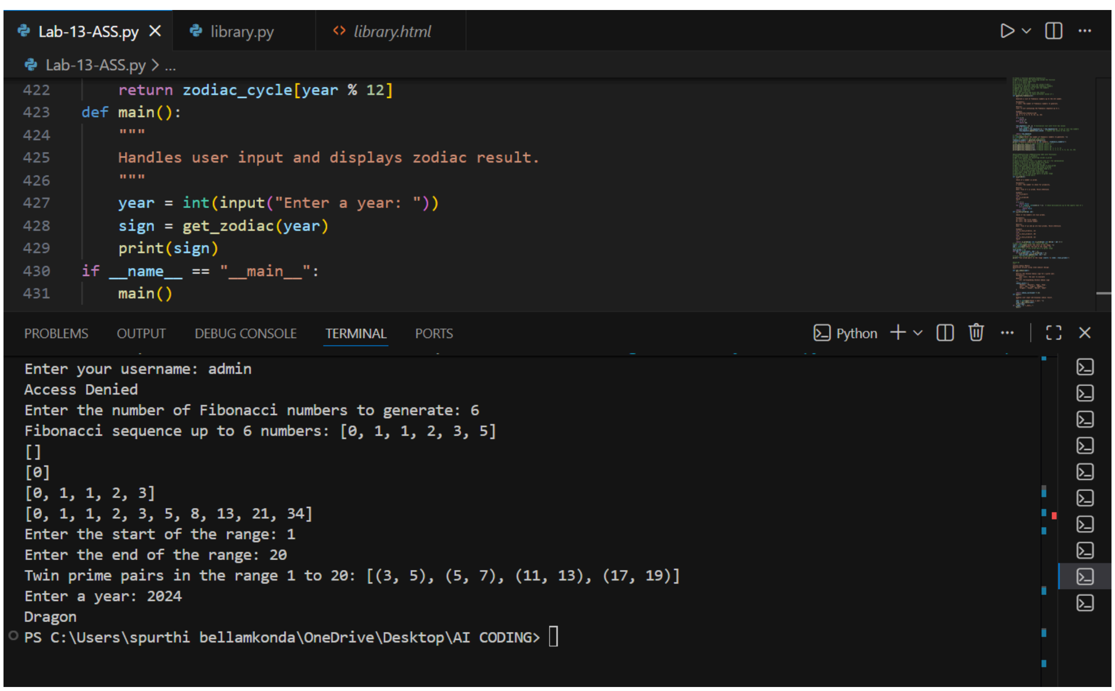
```
year = int(input("Enter a year: "))
if year % 12 == 0:
print("Monkey")
elif year % 12 == 1:
print("Rooster")
elif year % 12 == 2:
print("Dog")
elif year % 12 == 3:
print("Pig")
elif year % 12 == 4:
print("Rat")
elif year % 12 == 5:
print("Ox")
elif year % 12 == 6:
print("Tiger")
```

```
elif year % 12 == 7:
print("Rabbit")
elif year % 12 == 8:
print("Dragon")
elif year % 12 == 9:
print("Snake")
elif year % 12 == 10:
print("Horse")
elif year % 12 == 11:
print("Goat")
```

You must:

1. Create a reusable function: get_zodiac(year)
2. Replace the if-elif chain with a cleaner structure (e.g., list or dictionary).
3. Add proper docstrings.
4. Separate input handling from logic.
5. Improve readability and maintainability.
6. Ensure output remains correct.





Task 11 – Refactoring the Harshad (Niven) Number Checker

Refactor the given poorly structured Python script into a clean, modular, and reusable implementation.

A Harshad (Niven) number is a number that is divisible by the sum of its digits.**For example:**

- **18 → 1 + 8 = 9 → 18 ÷ 9 = 2 ✅ (Harshad Number)**
- **19 → 1 + 9 = 10 → 19 ÷ 10 ≠ integer ❌ (Not Harshad)**

Problem Statement

The current implementation:

- Mixes logic and input handling
- Uses redundant variables
- Does not use reusable functions properly
- Returns print statements instead of boolean values
- Lacks documentation

You must refactor the code to follow clean coding principles.

```
# Harshad Number Checker (Unstructured Version)
num = int(input("Enter a number: "))
temp = num
sum_digits = 0
while temp > 0:
digit = temp % 10
sum_digits = sum_digits + digit
temp = temp // 10
if sum_digits != 0:
if num % sum_digits == 0:
print("True")
else:
print("False")
else:
print("False")
```

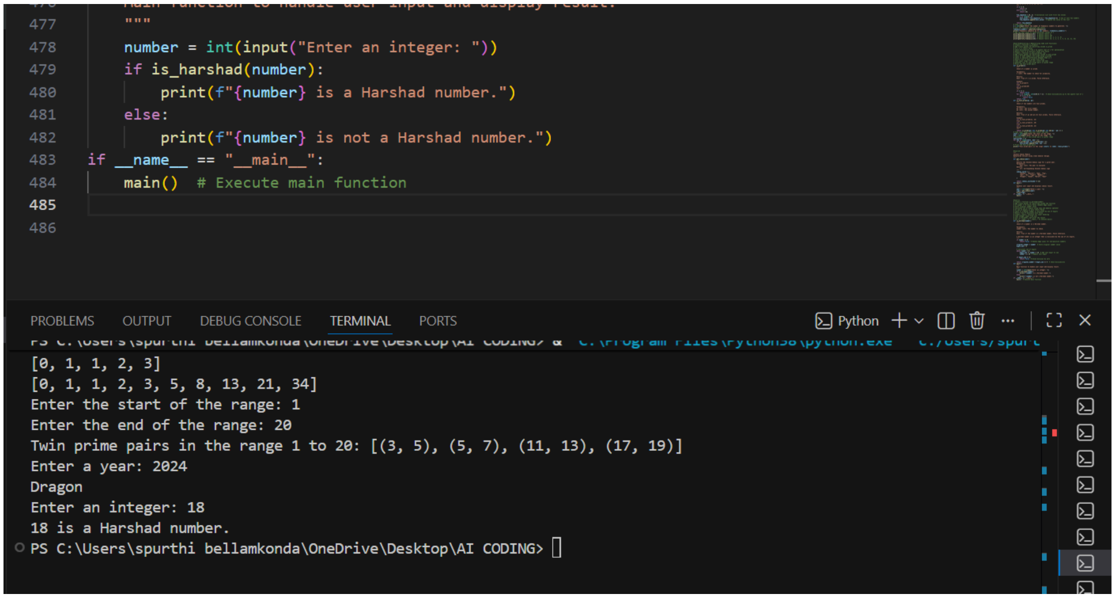
You must:

1. Create a reusable function: is_harshad(number)
2. The function must:
   - o Accept an integer parameter.
   - o Return True if the number is divisible by the sum of its digits.
   - o Return False otherwise.
3. Separate user input from core logic.

4. Add proper docstrings.
5. Improve readability and maintainability.
6. Ensure the program handles edge cases (e.g., 0, negative numbers).

```python
#Task-11
# Create a function is_harshad(number)
# Add triple quotes for docstring inside the function
# If number <= 0 return False (handle edge cases)
# Store original number value
# Calculate sum of digits using loop and modulus operator
# If sum of digits is zero return False
# Check if original number is divisible by sum of digits
# Return True if divisible else False
# Create a main() function for input handling
# Take integer input from user
# Call is_harshad() and print the result
# Use if __name__ == "__main__" to execute main()
def is_harshad(number):
    """
    Check if a number is a Harshad number.

    Parameters:
    number (int): The number to check.

    Returns:
    bool: True if the number is a Harshad number, False otherwise.

    A Harshad number is an integer that is divisible by the sum of its digits.
    """
```

```python
    if number <= 0:
        return False  # Handle edge cases for non-positive numbers

    original_number = number  # Store original number value
    digit_sum = 0

    # Calculate sum of digits
    while number > 0:
        digit_sum += number % 10  # Add last digit to sum
        number //= 10  # Remove last digit

    if digit_sum == 0:
        return False  # Avoid division by zero

    return original_number % digit_sum == 0  # Check divisibility
def main():
    """
    Main function to handle user input and display result.
    """
    number = int(input("Enter an integer: "))
    if is_harshad(number):
        print(f"{number} is a Harshad number.")
    else:
        print(f"{number} is not a Harshad number.")
```



Task 12 – Refactoring the Factorial Trailing Zeros Program

Refactor the given poorly structured Python script into a clean, modular, and efficient implementation.

The program calculates the number of trailing zeros in n! (factorial of n).

Problem Statement

The current implementation:

- Calculates the full factorial (inefficient for large n)
- Mixes input handling with business logic
- Uses print statements instead of return values
- Lacks modular structure and documentation

You must refactor the code to improve efficiency, readability, and maintainability.

 # Factorial Trailing Zeros (Unstructured Version)

n = int(input("Enter a number: "))

fact = 1

i = 1

while i <= n:

fact = fact * i

i = i + 1

count = 0

while fact % 10 == 0:

count = count + 1

fact = fact // 10

print("Trailing zeros:", count)

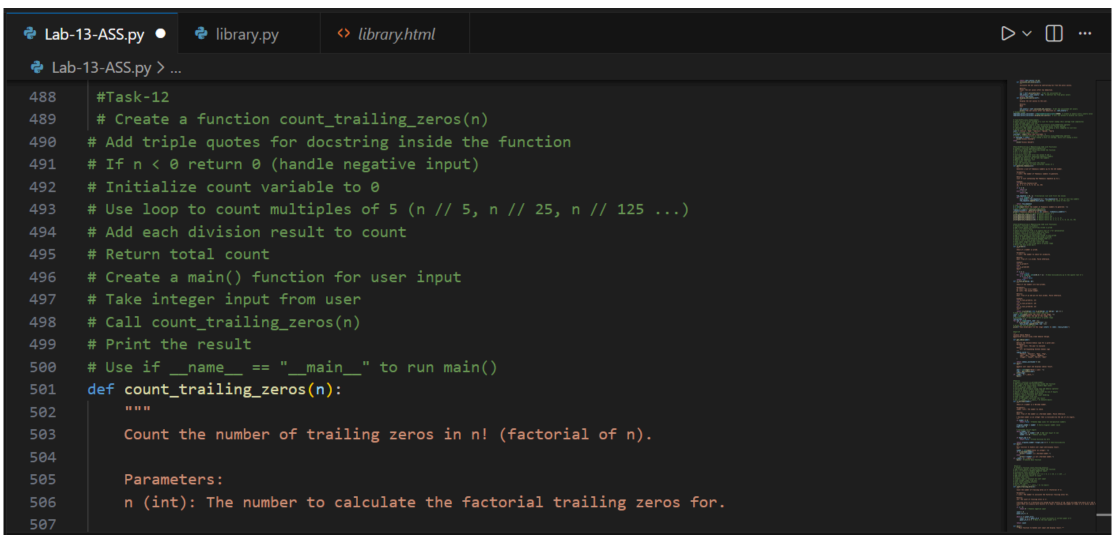
You must:

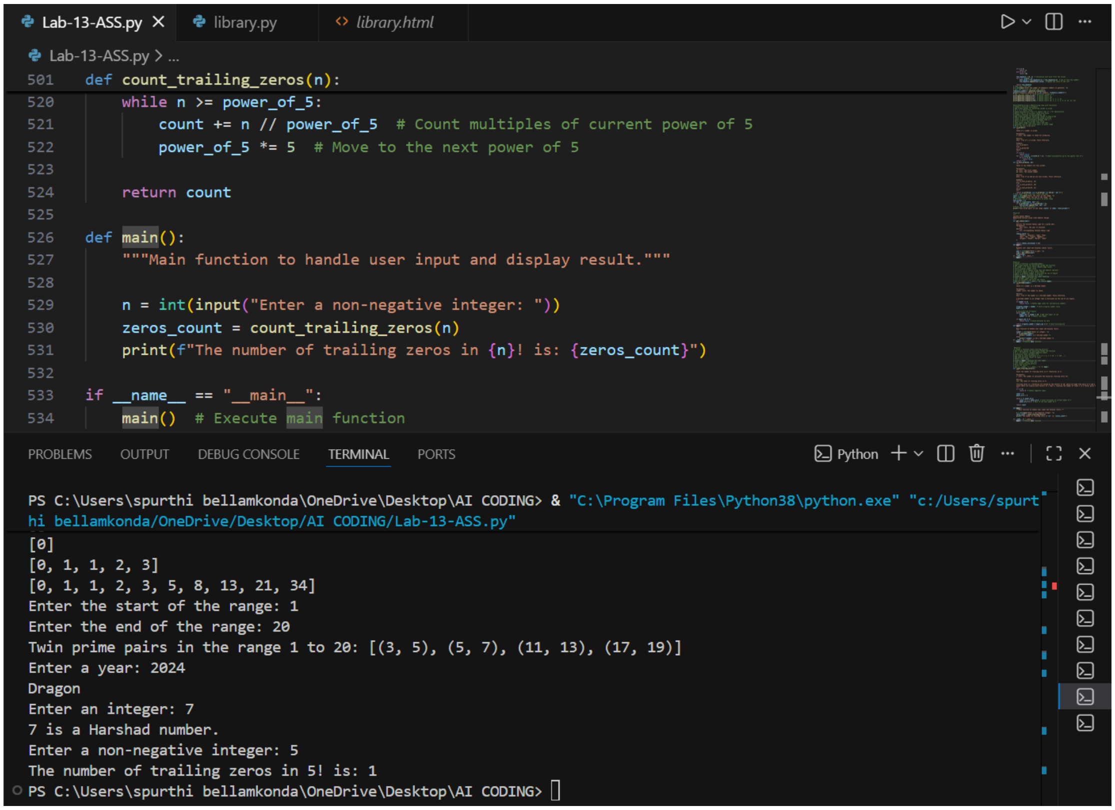
1. Create a reusable function: count_trailing_zeros(n)
2. The function must:

o Accept a non-negative integer n.

o Return the number of trailing zeros in **n!**.

3. Do NOT compute the full factorial.
4. Use an optimized mathematical approach (count multiples of 5).
5. Add proper docstrings.
6. Separate user interaction from core logic.
7. Handle edge cases (e.g., negative numbers, zero).





Task 13 (Collatz Sequence Generator – Test Case Design)

- Function: Generate Collatz sequence until reaching 1.Test Cases to Design:

· Normal: 6 → [6,3,10,5,16,8,4,2,1]

· Edge: 1 → [1]

· Negative: -5

· Large: 27 (well-known long sequence)

- Requirement: Validate correctness with pytest.

Explanation:

We need to write a function that:

· Takes an integer n as input.

· Generates the Collatz sequence (also called the 3n+1 sequence).

· The rules are:

o If n is even → next = n / 2.

o If n is odd → next = 3n + 1.

· Repeat until we reach 1.

- Return the full sequence as a list.

Example

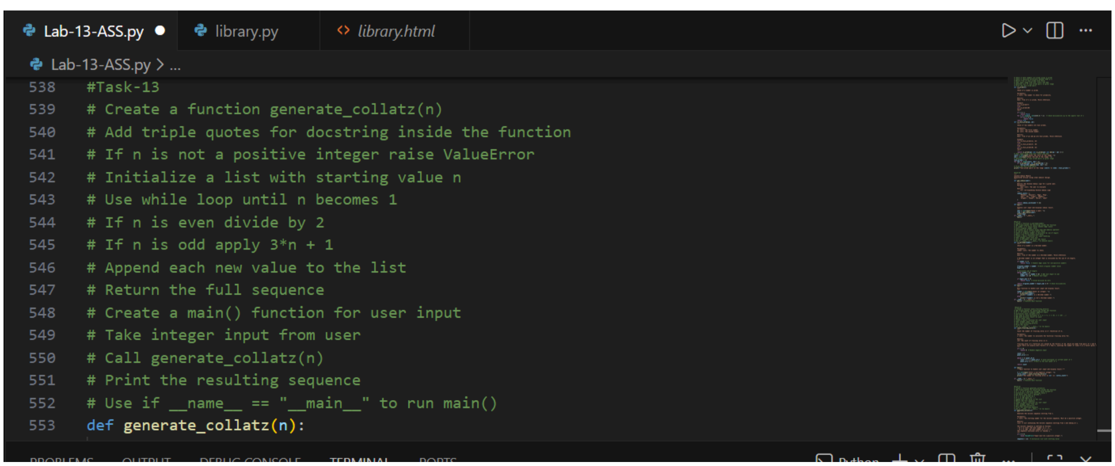
Input: 6

Steps:

- 6 (even → 6/2 = 3)
- 3 (odd → 3*3+1 = 10)
- 10 (even → 10/2 = 5)
- 5 (odd → 3*5+1 = 16)
- 16 (even → 16/2 = 8)
- 8 (even → 8/2 = 4)
- 4 (even → 4/2 = 2)
- 2 (even → 2/2 = 1)

Output:

[6, 3, 10, 5, 16, 8, 4, 2, 1]



Lab-13-ASS.py ● library.py library.html

Lab-13-ASS.py > ...

```python
def generate_collatz(n):
    """
    if n <= 0:
        raise ValueError("Input must be a positive integer.")

    sequence = [n]  # Initialize list with starting value

    while n != 1:
        if n % 2 == 0:  # If n is even
            n //= 2
        else:  # If n is odd
            n = 3 * n + 1
        sequence.append(n)  # Append each new value to the list

    return sequence
def main():
    """Main function to handle user input and display the Collatz sequence."""

    n = int(input("Enter a positive integer: "))
    try:
        collatz_sequence = generate_collatz(n)
        print(f"Collatz sequence starting from {n}: {collatz_sequence}")
    except ValueError as e:
        print(e)
if __name__ == "__main__":
    main()  # Execute main function
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS Python

```
Enter an integer: 8
8 is a Harshad number.
Enter a non-negative integer: 6
The number of trailing zeros in 6! is: 1
Enter a positive integer: 5
Collatz sequence starting from 5: [5, 16, 8, 4, 2, 1]
```

Task 14 (Lucas Number Sequence – Test Case Design)

- Function: Generate Lucas sequence up to n terms.(Starts with 2,1, then Fn = Fn-1 + Fn-2)Test Cases to Design:
- Normal: 5 → [2, 1, 3, 4, 7]
- Edge: 1 → [2]
- Negative: -5 → Error
- Large: 10 (last element = 76). Requirement: Validate correctness with pytest.

Lab-13-ASS.py ● | library.py | library.html

Lab-13-ASS.py > ...

```python
#Task-14
# Create a function generate_lucas(n)
# Add triple quotes for docstring inside the function
# If n is not a positive integer raise ValueError
# If n == 1 return [2]
# Initialize list with first two values [2, 1]
# Use loop to generate remaining Lucas numbers
# Append sum of last two numbers to the list
# Return the full Lucas sequence
# Add main block for manual testing
# Take integer input from user
# Call generate_lucas function
# Print the result
def generate_lucas(n):
    """
    Generate the Lucas sequence up to the nth number.

    Parameters:
    n (int): The number of Lucas numbers to generate. Must be a positive integer.

    Returns:
    list: A list containing the Lucas sequence up to n.

    The Lucas sequence is defined as follows:
    - L(0) = 2
    - L(1) = 1
```

Lab-13-ASS.py ● | library.py | library.html

Lab-13-ASS.py > ...

```python
def generate_lucas(n):
        raise ValueError("Input must be a positive integer.")

    if n == 1:
        return [2]  # Return the first Lucas number

    lucas_sequence = [2, 1]  # Initialize list with first two values

    for i in range(2, n):
        next_value = lucas_sequence[-1] + lucas_sequence[-2]  # Sum of last two numbers
        lucas_sequence.append(next_value)  # Append new value to the list

    return lucas_sequence
def main():
    """Main function to handle user input and display the Lucas sequence."""

    n = int(input("Enter the number of Lucas numbers to generate: "))
    try:
        lucas_numbers = generate_lucas(n)
        print(f"Lucas sequence up to {n} numbers: {lucas_numbers}")
    except ValueError as e:
        print(e)
if __name__ == "__main__":
    main()  # Execute main function
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
7 is a Harshad number.
Enter a non-negative integer: 5
The number of trailing zeros in 5! is: 1
Enter a positive integer: 3
Collatz sequence starting from 3: [3, 10, 5, 16, 8, 4, 2, 1]
Enter the number of Lucas numbers to generate: -5
Input must be a positive integer.
PS C:\Users\spurthi bellamkonda\OneDrive\Desktop\AI CODING>
```

Task 15 (Vowel & Consonant Counter – Test Case Design)

- Function: Count vowels and consonants in string.Test Cases to Design:
- Normal: "hello" → (2,3)
- Edge: "" → (0,0)
- Only vowels: "aeiou" → (5,0)

Large: Long text

- Requirement: Validate correctness with pytest.

```python
#Task-15
# Create a function count_vowels_consonants(text)
# Add triple quotes for docstring inside the function
# Initialize vowel and consonant counters to 0
# Convert text to lowercase for case-insensitive checking
# Loop through each character in the string
# If character is a vowel (a, e, i, o, u) increase vowel count
# Else if character is an alphabet letter increase consonant count
# Ignore digits, spaces, and special characters
# Return vowel and consonant counts as a tuple
# Add main block for manual testing
# Take string input from user
# Call the function and print the result
def count_vowels_consonants(text):
    """
    Count the number of vowels and consonants in a given text.

    Parameters:
    text (str): The input string to analyze.

    Returns:
    tuple: A tuple containing the count of vowels and consonants (vowel_count, consonant_count).

    This function counts the number of vowels (a, e, i, o, u) and consonants in the input text.
```

Lab-13-ASS.py ● | library.py | library.html

Lab-13-ASS.py > ...

```python
def count_vowels_consonants(text):
    vowel_count = 0
    consonant_count = 0
    vowels = "aeiou"

    for char in text.lower():  # Convert text to lowercase for case-insensitive checking
        if char in vowels:
            vowel_count += 1  # Increase vowel count
        elif char.isalpha():  # Check if character is an alphabet letter
            consonant_count += 1  # Increase consonant count

    return vowel_count, consonant_count  # Return counts as a tuple
def main():
    """Main function to handle user input and display vowel and consonant counts."""

    text = input("Enter a string: ")
    vowels, consonants = count_vowels_consonants(text)
    print(f"Number of vowels: {vowels}")
    print(f"Number of consonants: {consonants}")
if __name__ == "__main__":
    main()  # Execute main function
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
5 is a Harshad number.
Enter a non-negative integer: 6
The number of trailing zeros in 6! is: 1
Enter a positive integer: -7
Input must be a positive integer.
Enter the number of Lucas numbers to generate: -6
Input must be a positive integer.
Enter a string: hello
Number of vowels: 2
Number of consonants: 3
```